11.2 Vectors

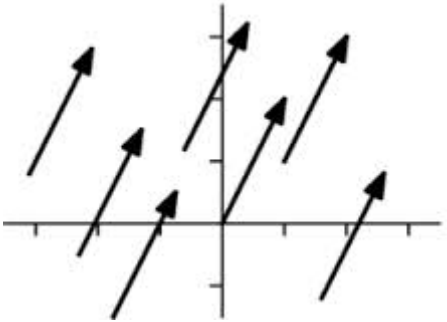
There are two types of quantities people measure: **Scalar** and **Vector**

Scalar
Determined by its magnitude only.
Example: length, temperature, voltage and mass

Vector
Determined by its magnitude and its direction, and usually represented by a directed line segment.
Example: velocity, acceleration and force

Definitions **(Vector, Initial and Terminal Point, Length)**

A **vector** in the plane is a **directed line segment**. The directed line segment $\vec{AB}$ has **initial point** A and **terminal point** B; its **length** is denoted by $|\vec{AB}|$. Two vectors are **equal** if they have the same length and direction.



Notation: $\mathbf{v}, \vec{v}, \overrightarrow{PQ} (\neq \overrightarrow{QP})$

Remark: A vector is measured by its length and direction, but not its position.

Given any directed line segment $\overrightarrow{PQ}$, we know there is a vector equal to $\overrightarrow{PQ}$ starting from the origin.

Definition:

If **v** is a **two-dimensional** vector in the plane equal to the vector with an initial point at the origin and terminal point (v_1, v_2), then the **component form** of **v** is $\mathbf{v} = \langle v_1, v_2 \rangle$

If **v** is a **three-dimensional** vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3), then the **component form** of **v** is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle$$

For example, $P(1, 2, 3)$ is a point in space, while $\overrightarrow{OP} = \langle 1, 2, 3 \rangle$ is a vector in space.

If $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is represented by the directed line segment $\overrightarrow{PQ}$, where $P(x_1, x_2, x_3)$ is the initial point and $Q(y_1, y_2, y_3)$ is the terminal point, then $v_i = y_i - x_i$ for $i = 1, 2, 3.$

Definition: **Standard position vector** is the directed line segment with the initial point at the origin.

For example, consider the points $A(1, 1, 1)$ and $B(1, 2, 3)$ with the vector

$$\overrightarrow{AB} =< 0, 1, 2 >= \overrightarrow{OP}$$

Here P is the point $P(0, 1, 2)$ and $\overrightarrow{OP}$ is the standard position vector that is equal to $\overrightarrow{AB}$.

Definition: The **magnitude** or **length** of the vector **v** = $\overrightarrow{PQ}$

If we have two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, then

$$|\overrightarrow{PQ}| = |\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}|$$

Example: Find the component form and length of the vector with initial point $P(-3, 4, 1)$ and the terminal point $Q(-5, 2, 2)$.

Vector Algebra Operations

Two principal operations involving vectors are *vector addition* and *scalar multiplication*. A **scalar** is simply a real number.

Definition **Vector Addition and Multiplication of a Vector by a Scalar**

Let $\vec{u} =< u_1, u_2, u_3 >$, $\vec{v} =< v_1, v_2, v_3 >$, $k \in \mathbb{R}$.

Addition:

Scalar Multiplications:

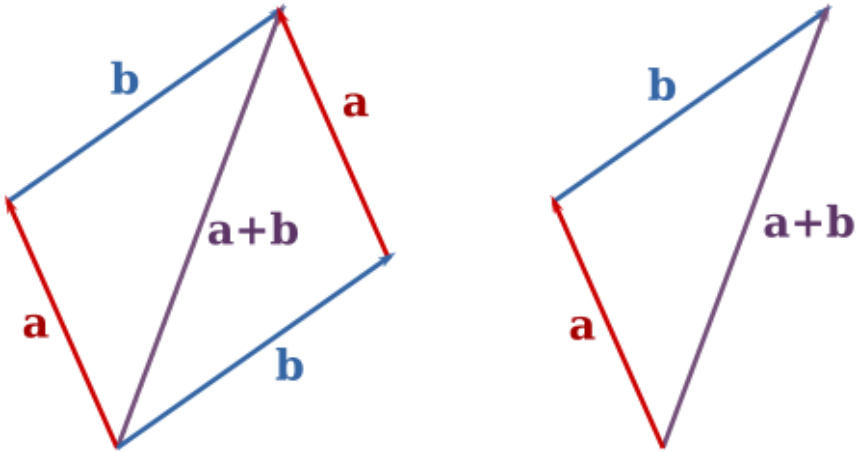
Vector Difference:

Vector Addition

Given two vectors **a** and **b**, the vector addition **a** + **b** can be illustrated geometrically in two ways.

1. **Parallelogram Law**
   If **a** and **b** are represented by two sides of a parallelogram, then **a** + **b** is a vector given by the diagonal of the parallelogram.

2. **Triangle Law**
   If **a** and **b** are represented by two sides of a triangle, then **a** + **b** is a vector given by the third side of the triangle.



Scalar Multiplication

Let c be a scalar and **v** a nonzero vector. Then

1. $|c\mathbf{v}| = |c||\mathbf{v}|$
2. Two vectors are parallel ⇔ one is scalar multiple of the others. That is, **u**||**v** ⇔ **u**= c **v** for some scalar c. In addition, u and v are in the same direction if $c > 0$; and in the opposite direction if $c < 0$.

Example:

1. $\mathbf{u} =< 3, 5, 4 >, \mathbf{v} =< -\frac{3}{2}, -\frac{5}{2}, -2 >$

2. $\mathbf{u} =< 3, 5, 4 >, \mathbf{w} =< 3, 6, 0 >$

Example Performing Operations on Vectors

Let $\mathbf{u}=\langle -1, 3, 1 \rangle$ and $\mathbf{v}=\langle 4, 7, 0 \rangle$.

(a). $2\mathbf{u}+3\mathbf{v}$

(b). $\mathbf{u}-\mathbf{v}$

(c). $\left|\frac{1}{2}\mathbf{u}\right|$

Properties of Vector Operations

Vector operations have many of the properties of ordinary arithmetic.

Let **u**, **v**, **w** be vectors and $a,\ b$ be scalars.

1. $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$
2. $(\mathbf{u}+\mathbf{v})\ +\mathbf{w}=\mathbf{u}+\ (\mathbf{v}+\mathbf{w})$
3. $\mathbf{u}+\mathbf{0}=\mathbf{u}$
4. $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$
5. $0\mathbf{u}=\mathbf{0}$
6. $1\mathbf{u}=\mathbf{u}$
7. $a(b\mathbf{u})=(ab)\mathbf{u}$
8. $a(\mathbf{u}+\mathbf{v})=a\mathbf{u}+a\mathbf{v}$
9. $(a\ +\ b)\mathbf{u}=a\mathbf{u}+b\mathbf{u}$

Unit Vectors

A vector **v** of length 1 is called a unit vector. The *standard unit vectors* are

$$\mathbf{i} = \langle 1, 0, 0 \rangle$$
$$\mathbf{j} = \langle 0, 1, 0 \rangle$$
$$\mathbf{k} = \langle 0, 0, 1 \rangle$$

Any vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ can be written as a linear combination of the standard unit vectors as follows:

Remark:
We can express any nonzero vector in terms of its two important features, its length and direction.

Given $\mathbf{v} \neq \mathbf{0}$, its length is $|\mathbf{v}|$ and unit vector in its direction is $\frac{1}{|\mathbf{v}|}\mathbf{v}$. So we can write

$$\mathbf{v} = |\mathbf{v}| \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right).$$

Example: Find a unit vector **u** in the direction of the vector from $P_1(1, 0, 1)$ to $P_2(3, 2, 0)$.

Example:

1. Find the position vectors from $P(2, 3, 1)$ and $Q(-\ 3, 0, 0)$. Find a unit vector in the direction of the vector from P to Q.

2. Which of the following are unit vectors?

a. $\mathbf{u} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$

b. $\mathbf{v} = \frac{1}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}$

c. $\mathbf{w} = \sin\theta\mathbf{i} - \cos\theta\mathbf{k}$

3. Find a vector of magnitude 3 in the direction of the vector $\mathbf{v} = \frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} - \mathbf{k}$.

The midpoint of a Line Segment

Suppose we have two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in space. Let M be the midpoint of $\overrightarrow{P_1P_2}$.

